

■ SQL Module 3 - Lab 1: LIKE, ORDER BY, GROUP BY (Cheat Sheet)

Welcome to your **Intermediate SQL Revision Notes!** In this lab, you leveled up your SQL skills with pattern matching, sorting, and grouping techniques. Let's break them down in a fun and easy way! ■

■ LIKE Operator

- 1 Use ****%**** to represent zero or more characters.
- 2 Use ****_**** (underscore) to represent a single character.
- 3 It's used in the ****WHERE**** clause for pattern-based filtering.

■ Example:

```
SELECT f_name, l_name FROM employees WHERE address LIKE '%Elgin,IL%';
```

→ Finds all employees whose address contains 'Elgin,IL'.

■ BETWEEN Operator

The BETWEEN operator selects values within a given range (inclusive).

■ Example:

```
SELECT * FROM employees WHERE salary BETWEEN 40000 AND 80000;
```

→ Returns employees earning between 40,000 and 80,000.

■ ORDER BY Clause

- 1 Default sorting is ascending (A–Z or smallest to largest).
- 2 You can sort by multiple columns using commas.
- 3 Helpful for organizing your data neatly!

■ Example:

```
SELECT f_name, l_name, dep_id FROM employees ORDER BY dep_id DESC, l_name;
```

→ Sorts by department (descending) and then alphabetically by last name.

■ GROUP BY Clause

- 1 Used with aggregate functions like COUNT(), SUM(), AVG(), etc.

2 Groups data by one or more columns with identical values.

■ Example:

```
SELECT dep_id, COUNT(*) FROM employees GROUP BY dep_id;
```

→ Returns department IDs with the count of employees in each.

■ HAVING Clause

HAVING filters grouped data — similar to WHERE but used *after* grouping.

■ Example:

```
SELECT dep_id, COUNT(*) AS "NUM_EMPLOYEES", AVG(salary) AS  
"AVG_SALARY" FROM employees GROUP BY dep_id HAVING COUNT(*) < 4  
ORDER BY AVG_SALARY;
```

→ Filters departments with fewer than 4 employees and orders by average salary.

■ Quick Summary

- 1 LIKE → For pattern matching in text.
- 2 BETWEEN → For selecting ranges of values.
- 3 ORDER BY → For sorting results.
- 4 GROUP BY → For grouping similar data.
- 5 HAVING → For filtering grouped data.

👤 ■ Authors: Lakshmi Holla, Pratiksha Verma, Abhishek Gagneja

■ You've mastered the art of grouping, sorting, and filtering like a pro! Keep practicing to become a true SQL Jedi! ■